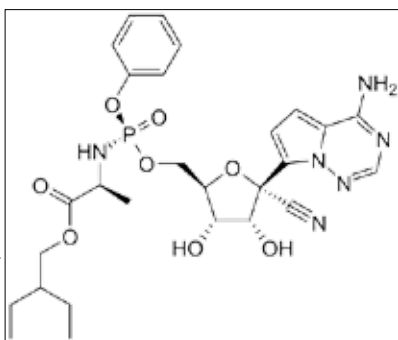


Source: Wikipedia



Structure of Remdesivir which is currently being developed by Indian pharma companies

clinical trial was carried out in China with 237 critically-ill COVID-19 patients. 158 patients received Remdesivir while 79 got the placebo. The death rate, however, was similar in both groups – 13.9 percent for the Remdesivir-administered patients and 12.8 percent in the placebo group. The post was allegedly put up on the WHO website and immediately removed, however, a medical news website STAT captured a screenshot of the inadvertent post. Gilead Sciences said in a statement that the results were prematurely posted and the post included “inappropriate characterizations of the study”. The company believes that the study was terminated early due to low enrollment and therefore, it is not qualified to provide meaningful conclusions.

Hydroxychloroquine

Unfortunately, the anti-malaria drug, hydroxychloroquine that India was banking on the most since it was touted to be a potential cure for COVID-19, failed in a clinical study. The study showed that the drug had no additional benefit in fighting against the disease over standard care. In fact, treatment using hydroxychloroquine recorded more deaths. The death rates for patients on hydroxychloroquine were 28 percent, when combined with antibiotic azithromycin the death rates were 22 percent and finally, the death rates for those on standard care were 11 percent, as per the study. However, this US-funded analysis has not yet been peer-reviewed and it does have some limitations according to experts, so all hope is not lost.

India is one of the largest producers of hydroxychloroquine and recently supplied the anti-malarial drug to 55 COVID-19-hit countries including the US, Afghanistan, Bhutan, Nepal, Maldives, Uganda, Netherlands, France, Ukraine, and many others. The FDA has advised against the purchase and use of the anti-malarial drug without a prescription. And, for good reason. Patients with heart ailments or other underlying diseases can become critically-ill if the drug is taken without being prescribed or administered by healthcare professionals. It is also strictly prohibited for those below 15 years and above 60 years due to possible side-effects.

Tocilizumab therapy

Another possible path-breaking development in the treatment of COVID-19 is tocilizumab (TCZ) therapy that has been used by doctors at Indore's Sri Aurobindo Institute of Medical Sciences (SAIMS). TCZ is an immunosuppressant primarily used to treat rheumatoid arthritis. Dr Ravi Dosi, a pulmonologist leading a 100-member team of doctors, nurses and paramedical staff at the hospital stated that they administered the tocilizumab therapy to five critical patients, two of whom made a drastic recovery. This therapy was administered in those cases where the conventional treatment – doses of hydroxychloroquine, steroids, oseltamivir, antibiotics and ventilator support – had failed. The therapy is used against a phenomenon called cytokine storm which primarily proves to be fatal for COVID-19 patients. Cytokine is a protein secreted by the immune system to combat infection, however, excess secretion can lead to a



TCZ (tocilizumab) therapy has been used to treat 'cytokine storm' in COVID-19 patients

'cytokine storm' which causes deathly hyper inflammation in the body.

Other experimental treatments

Mumbai-based Cipla, along with Remdesivir, is also working towards synthesising Japanese flu drug Favipiravir, which is also strictly for R&D and cannot be marketed since it is patented. Another drug being investigated is Kaletra, a concoction of two antiviral medicines – lopinavir and ritonavir. This drug was previously used to treat HIV patients and is now being administered in some cases in India to critically-ill or old hospitalised patients.



Kaletra, an HIV drug, is also being looked into as a treatment option

A British National in Kerala who tested positive for COVID-19 was given Kaletra and he has recovered since. The patient was administered the experimental drug for a period of seven days at Ernakulam Medical College. Another possible treatment plan which has been used by healthcare professionals in India is a concoction of HIV, swine flu and Malaria drugs. This combination was used in March by the doctors at Sawai Man Singh Hospital in Jaipur to treat an Italian citizen who travelled to India and tested positive for COVID-19. A combination of 200mg Lopinavir and 50mg Ritonavir (used for HIV treatment), chloroquine (used for the treatment of Malaria) and oseltamivir (used for the treatment of swine flu) were administered to the Italian woman. Following the novel treatment, the woman, shortly after, tested negative for COVID-19. 